Honest Baselines at National Scale: Leakage-Resistant Distributed Forecasting of Indian Mandi Prices with Conformal Intervals and a Multilingual Farmer Interface

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Abstract

Most studies of Indian agricultural price forecasting model a few markets, use a single proportional hold-out, and omit naive baselines and prediction intervals. We built AG-DATA, a Scala/Spark/Kafka/HDFS system, and used it to benchmark mandi-level modal-price forecasts at national scale. The benchmark covers 11,845 market–commodity–variety series for 36 staple commodities. It has four horizons (next report, 3, 7 and 14 days) and uses chronological, target-purged train (2016–2023), validation (2024) and test (2025–April 2026) periods. Five naive and seasonal baselines are compared with Spark MLlib learners. Three findings stand out. First, 17–34% of target price changes are exactly zero. As a result, squared-loss learners do not beat persistence under absolute error, while absolute-loss gradient boosting does. Second, the gains of the validation-selected models over persistence are modest but significant and broad: MAE is 2.9%, 4.7%, 6.2% and 7.7% lower at the four horizons (Diebold–Mariano $p < 10^{-8}$), with improvements on 58–76% of series and on all 36 commodities at 7 days. A per-series exponential smoother matches the learned models at 1–3 days. Third, split-conformal intervals with volatility bins reach 78.6–80.5% test coverage at a nominal 80%, but only 68–73% after a change in the publishing system in November 2025. That change also fragments series identifiers unless they are harmonised. The forecasts are served through a read-only API to an installable, offline-capable web application in six Indian languages. The application presents cautious SELL/HOLD/WATCH/COMPARE market-outlook cards. Their rule abstains on about 98% of report-days and is back-tested on the test period. We release the code, tests and experiment manifests, and we discuss why published accuracy gains over ARIMA are not comparable with these results.

Keywords: agricultural price forecasting; AGMARKNET; mandi; Apache Spark; conformal prediction; forecast evaluation; decision support; multilingual mobile application

1 Introduction

Indian farmers sell most of their produce through regulated wholesale markets (*mandis*). Each mandi publishes a daily minimum, maximum and modal (most frequent) price per commodity and variety through AGMARKNET, and these reports are mirrored on the Open Government Data platform. The reports cover thousands of markets and hundreds of commodities. They are the most detailed public price signal available to a farmer deciding whether to sell today, wait, or take produce to another market.

Forecasting these prices has a large literature (Section 2). Most of it studies a single market, a few markets of one state, or monthly national indices, with deep or hybrid models evaluated on a single proportional hold-out. Naive baselines are rarely reported, and prediction intervals are almost never reported [16, 20, 25, 26]. Forecast-evaluation research warns that such protocols can overstate gains: simple benchmarks are hard to beat [17, 18], scale-free errors are needed when series differ in level [10], and leakage through splitting and preprocessing is a common pitfall [3, 9]. Studies of market-information services in India also show that price information alone often does not raise the prices farmers receive [5, 23], although some interventions have had large effects [8, 12]. A farmer-facing forecast should therefore be honest about its uncertainty and should not be presented as a trading signal.

This paper reports the design and evaluation of AG-DATA, a distributed system built on Scala, Spark, Kafka and HDFS that forecasts mandi-level modal prices for 36 commodities and serves the forecasts, with conformal intervals whose coverage we measure and with cautious decision cards, to an installable six-language web application. Our contributions are evidence-based and deliberately modest:

1. **A leakage-resistant benchmark at national scale.** The benchmark spans 11,845 mandi–commodity–variety series and 21.1M report-level rows. It uses chronological train/validation/test periods with target-date purging, five naive and seasonal baselines, and four horizons. Results are reported with MASE, per-horizon Diebold–Mariano tests and per-series win rates (Sections 6–8).

2. **An empirical finding about the loss function.** Between 17% and 34% of target price changes are *exactly* zero, because mandis often re-quote an unchanged modal price. This makes the choice of training loss decisive under absolute-error criteria. At the next-report horizon, squared-loss ridge regression and random forests do not differ significantly from persistence, whereas absolute-loss gradient boosting is 2.9% better. A per-series exponential smoother is as good as any learned model at 1–3 days (Section 8).

3. **Measured, not assumed, uncertainty.** Split-conformal intervals with Mondrian volatility bins are calibrated on a validation year. Their test-period coverage is reported by horizon, by volatility bin and before and after a change in the publishing system (Section 6).

4. **Data-integrity findings that affect any study of this source.** These are a grade relabel in January 2025 and a market rename and volume collapse after the November 2025 migration of AGMARKNET. Left untreated, they fragment series and silently shrink the test set (Section 4).

5. **An end-to-end, reproducible system.** The system has shared batch/stream canonicalisation that is tested for parity, a Kafka/Structured Streaming path with stated delivery semantics, an HDFS layout, an API, and a low-bandwidth progressive web app in six languages whose decision rule is back-tested (Sections 7–9).

We do not claim superiority over published models: no prior study uses the same data, horizons and splits, and Section 8.10 states why each published number is context only.

2 Related work

Indian agricultural price forecasting. Most work comes from statistical and machine-learning groups modelling a few series in depth.

- Paul et al. compared ARIMA, GRNN, SVR, random forests and gradient boosting on daily AGMARKNET brinjal prices from 17 Odisha markets, with Diebold–Mariano tests and a model confidence set [25].
- A later study of monthly wholesale prices for 19 commodities in 143 markets found deep recurrent models ahead of ML and stochastic models. Unusually, it included a random-walk-with-drift baseline [26].
- Wavelet hybrids, time-delay networks and ANN/linear hybrids have been proposed for oilseeds, pulses and other commodities [1, 14, 15, 24].
- ARMA–GARCH models have been fitted to a single onion market [16], and LSTM variants to potato and other prices [11, 19, 27].
- The closest study to ours in data coverage is Manogna et al. [20]: 23 commodities, 165 markets, daily AGMARKNET data, 2010–2024. It uses an 80/20 split with sliding windows, reports RMSE and MAPE, and explicitly decides against MASE. Its abstract and body give different MAPE values for the same models.
- National CPI sub-indices for tomato, onion and potato have been forecast with autoregressive models [30].
- Reviews summarise the shift from statistical to intelligent and combined models [29].

Big-data systems. Spark [32] and MLlib [22] are general platforms. The nearest agricultural precedent we found uses Spark to clean multi-city prices in Pakistan and then models a single series [28]. We found no system that trains, evaluates, calibrates and serves forecasts for thousands of mandi series within one distributed pipeline.

Volatility and anomaly detection. The IFPRI early-warning system flags excessive price variability in international futures with nonparametric thresholds on returns [6]. LSTM–GARCH hybrids have modelled volatility in Indian prices [21]. We found no mandi-level system that combines point forecasts, intervals and anomaly flags.

Evaluation methodology and uncertainty. Our protocol draws on the following:

- scaled errors [10] and the Diebold–Mariano test [4];
- the M4 and M5 competitions, which show that simple methods are strong [17, 18];
- work on when cross-validation is valid for time series and on how to estimate performance [2, 3];
- a catalogue of evaluation pitfalls [9].

Conformal methods for time series include EnbPI [31], ensemble conformalised quantile regression [13] and adaptive conformal inference under shift [7]. We use the simplest member of this family, split conformal with Mondrian bins, and *measure* its coverage.

Market information and farmer welfare.

- Mobile phones reduced price dispersion among Kerala fishermen [12].
- Internet kiosks raised soy prices in Madhya Pradesh [8].
- Randomised SMS price information in Maharashtra had small effects [5].
- Wholesale-price information for West Bengal potato farmers did not change average farm-gate prices [23].

These results motivate our framing of decision cards as cautious market outlooks.

Gap. To our knowledge, none of the reviewed Indian studies jointly does all three of the following: (i) evaluates daily mandi-level modal prices across thousands of market series; (ii) uses strictly chronological, purged splits with naive and seasonal baselines and scale-free errors at several horizons; (iii) reports distribution-free intervals whose coverage is measured out of sample. AG-DATA is designed to test whether doing all three changes the conclusions drawn from smaller studies.

3 Problem formulation

An *entity* e is a price series identified by (state, district, market, commodity, variety). Grade is excluded from the key for the reason given in Section 4. Entity e publishes modal prices $P_{e,t}$ (INR per quintal) on irregular report dates t. For an origin t at which e reports, and a horizon $h \in \{3, 7, 14\}$ days, the target is the first report of e dated in $[t+h,\, t+h+3]$. If no report falls in that window, the origin has no label for h. The *next* horizon targets the next report, provided it is at most 7 days away. Models predict the log return $y_{e,t,h} = \ln P_{e,\tau} - \ln P_{e,t}$, where τ is the target date. Forecasts in rupees are $\hat{P} = P_{e,t} \exp(\hat{y})$, and every error metric is computed on that scale. Using a log-return target lets one global model serve commodities whose price levels differ by two orders of magnitude.

Three tasks are kept separate:

- **forecasting**, the subject of this paper's main results;
- **anomaly detection**, which flags reports that deviate from the entity's trailing behaviour;
- **decision support**, a transparent rule on top of forecast intervals.

None of these models causal effects.

4 Data and provenance

Source. The primary source is an archive of AGMARKNET daily reports for 2001-01-10 to 2026-04-21, distributed as a Kaggle dataset (`khandelwalmanas/daily-commodity-prices-india`, version 134, published under GODL-India). Our local copy matches that release file by file in byte size; content hashes were not compared. Its columns match the official data.gov.in resource *Variety-wise Daily Market Prices Data of Commodity*. The live data.gov.in API (resource `9ef84268-...`) needs a key, which we did not have; its adapter is implemented and tested offline. The archive has 75,984,017 rows. After quality classification and business-key de-duplication, the silver layer has 75,644,801 rows; 97.6% of these are internally consistent (`VALID`).

Scope. We use 36 staple commodities: vegetables, fruits, cereals, pulses, oilseeds and cotton (33,116,234 valid reports since October 2014). They were chosen from training-period data with four steps:

1. take an a-priori list of staple crop-name patterns;
2. keep matches with more than 50,000 valid reports in 2021–2023;
3. drop two duplicate or processed forms;
4. keep the 36 most-reported.

This is not the overall top 36: six high-volume items outside the pattern list, such as radish, papaya and jaggery, are excluded.

Identifier harmonisation. Two changes in how the source publishes identifiers would otherwise break series:

- **Grade relabel.** From 2025-01-27 many series switch their grade label from *FAQ* to *Local*, with no overlap. Across all (state, district, market, commodity, variety, date) keys since 2023, none carries two grades on the same day, so we treat grade as an attribute.
- **Market rename.** With the AGMARKNET 2.0 migration (portal live 2025-11-07), 95–97% of reports in December 2025 to March 2026 (77% in April) carry an " APMC" suffix on the market name, which we strip.

Harmonisation merges 117,623 fragmented series into 75,814 entities. It removes 191 same-day collisions. Without it, essentially no panel series would continue into 2026. Around the migration, monthly volume collapses to about 6% of its earlier level in November 2025 and recovers only to roughly 20–35% in December 2025 to April 2026, so we report test metrics separately before and after November 2025.

Cleaning. A causal sanity rule removes reports that differ by more than a factor of five from the geometric mean of the entity's previous ten reports. These are typically unit or keying slips. The rule removes 28,567 reports (0.086%). Genuine price spikes below that factor are kept.

Panel. An entity enters the training and evaluation panel if it has at least 300 report days in 2021–2023. This rule uses training-period information only. The panel has 11,845 entities and 21.06M feature rows. Thinly reported entities still receive forecasts in the application, with a warning.

5 Forecasting methodology

Features. Each row is a report of entity e at origin t and carries 24 base features plus one horizon-specific feature, all computed from reports dated $\leq t$ (Table 1). Missing values are set to zero. Every feature that can be missing has a companion indicator or count (`has_ly`, `cnt30`), so no fitted imputation statistic is needed and none can leak.

Baselines. Five baselines are reported, all as log returns:

- persistence ($\hat{y} = 0$);
- geometric moving averages over 7 and 30 days;
- seasonal naive (last year's level around $t + h - 365$; persistence when unavailable);
- seasonal-adjusted persistence (today's level times last year's change over the same horizon).

Table 1: Feature groups ($\ell = \ln P$). All frames end at the origin.

Group	Features
Level / lags	ℓ_t; $\ell_t - \ell_{t(-k)}$ for $k \in \{1, 2, 5\}$ previous reports; days since previous report
Rolling (7/30/90 d)	deviation from the rolling mean; std (30, 90); report counts (30, 90); distance to 30-day max/min
Range	$(\max - \min)$/modal on day t
Seasonal	deviation from the level at $t - 365 \pm 3$; indicator; last year's change from $t - 365$ to $t + h - 365$
Cross-section	same-day mean 30-day deviation and 1-report return of the commodity nationally and within the state; number of reporting markets
Calendar	day-of-year (sine, cosine); day of week
Categorical	commodity index (fitted on training rows)

Learners. All learners are Spark MLlib models, one per horizon (direct strategy):

- ridge regression with squared and Huber losses;
- a random forest (60 trees, depth 10);
- gradient-boosted trees with squared loss (depth 6, 200 iterations, step 0.05);
- gradient-boosted trees with absolute loss in three configurations.

MLlib's absolute-loss boosting fits each tree after the first to sign of the residual, without a leaf line search. Each iteration therefore moves the prediction by up to the step size, which is large relative to typical log returns of about 0.01. We therefore train these models on the target in percent with step sizes of 0.25–0.5, and rescale inside the persisted pipeline. A unit test confirms that this pipeline recovers the conditional median of a zero-inflated target, whereas squared loss recovers the mean.

6 Experimental protocol

Periods.

- Training origins: 2016-01-01 to 2023-12-31.
- Validation origins: 2024.
- Test origins: 2025-01-01 to 2026-04-21.

A row is kept in a period only if its target date also lies in that period. This purge removes the one path by which a label can cross a boundary, and a unit test checks it. Training uses a uniform random sample of 2×10^6 rows per horizon (seed fixed), drawn from 15–16M eligible rows. Hyper-parameters are tuned on a random sample of 6×10^5 validation rows.

Selection and amendments. For each horizon, the family with the lowest validation MAE (in rupees) is selected, baselines included. The search space was amended twice, before any test metric was read, on validation evidence only:

- first, because squared-loss learners tied persistence at the next-report horizon, where 34.4% of target changes are exactly zero;
- second, because MLlib's absolute loss is mis-scaled for log returns, as described above.

The logs of both aborted runs are kept in the repository.

Metrics. The following are computed on the rupee scale over all test rows:

- MAE, RMSE, WAPE $= \sum |e| / \sum |A|$, sMAPE, and bias (mean signed error);
- MASE, scaled by each entity's in-sample h-step naive MAE on training rows [10];
- relative MAE against persistence on identical rows;

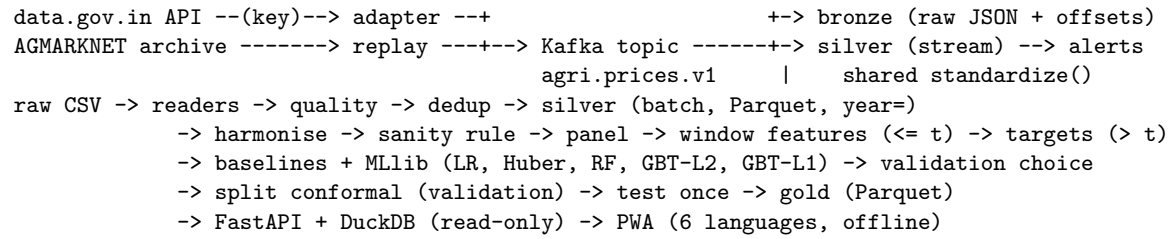


Figure 1: AG-DATA data flow. Historical data takes the batch path; Kafka carries only incremental reports.

- directional accuracy on rows with $|y| \geq 1\%$.

MAPE is not used: low-priced leafy vegetables make its denominators unstable. Diebold–Mariano tests [4] compare each model with persistence on the series of daily mean absolute-error differences, using a Newey–West variance (lag $= h$) and the Harvey–Leybourne–Newbold correction. Errors are aggregated per date first because same-day errors are strongly correlated across markets. We also report the share of series (those with at least 30 test rows) on which the selected model beats persistence.

Intervals. For the selected model we use split-conformal intervals with Mondrian bins: validation residuals $r = y - \hat{y}$ are grouped into five volatility bins by 30-day log-price standard deviation, with bin edges taken from validation quantiles. The 5/10/90/95% residual quantiles of each bin give 80% and 90% intervals. Exchangeability does not hold for time series, so coverage is measured on the test period rather than assumed [7, 31].

Stability. A rolling-origin backtest refits the selected configuration on expanding windows ending in 2020–2023 and tests on each following year.

Anomaly detection. No labels exist for real anomalies. Instead, corruptions are injected into 1% of 400,000 sampled 2025 test rows, half of each kind:

- unit or keying errors ($\times 10$, $\times 100$, $\div 10$);
- shocks of ± 35–60%.

We compare the production rule (trailing robust z beyond 3.5 with a move of more than 20%, or more than a factor of five) with two baselines: a previous-report change and the legacy commodity-IQR rule, refitted on training data.

7 System design

Figure 1 shows the data flow. Each component exists for a stated reason; the full rationale is in the repository's `docs/ARCHITECTURE.md`.

Batch layer. Scala jobs on Spark 4.2 DataFrames read the raw CSVs with explicit schemas, so no schema inference is needed. They classify each record against seven quality rules and de-duplicate on a seven-column business key with grouped aggregation rather than a whole-row `distinct`, which had run out of memory in an earlier version. The output is a year-partitioned Parquet silver layer.

Feature pipeline. Features are computed with window functions partitioned by entity and ordered by an integer day number:

- 5 level, lag and return features;
- 10 rolling and range features over 7/30/90 calendar days, using `rangeBetween`;
- 3 seasonal features at $t - 365$ and the same-horizon change one year earlier;
- 4 same-day cross-sectional aggregates for the commodity, nationally and within the state;
- 3 calendar terms;

- the commodity index.

Every feature frame ends at offset 0 and every target frame starts at offset $+h \geq 1$. A perturbation test multiplies all future prices, and all prices of another entity, by arbitrary factors. It asserts that no feature at the origin changes, so any forward-looking or cross-entity window fails the test. The job stages intermediate results as Parquet. An earlier version persisted 33M-row frames with `DISK_ONLY` caching and exhausted the development machine's disk.

Models. MLlib pipelines wrap each candidate: ridge regression and Huber regression on standardised features with one-hot commodity, and random forests and gradient-boosted trees on raw features with an indexed commodity. The indexer and scaler are fitted on training rows only. Fitted pipelines are persisted per experiment and applied unchanged when forecasts are published, so serving needs no separate model runtime.

Streaming layer. A Python adapter publishes canonical JSON events to a six-partition Kafka topic. The adapter has retry with jittered exponential backoff, client-side rate limiting, atomic checkpoints and key redaction. The message key is the business key, and the event id is a content hash. Spark Structured Streaming writes two outputs:

- a *bronze* file sink with every record and its offsets;
- a *silver* `foreachBatch` sink.

The silver path parses, quarantines unknown schema versions and malformed messages, and canonicalises with the same function as the batch layer. It then drops exact re-sends by event id within a 10-day event-time watermark and scores alerts against batch-computed trailing statistics. Writing each micro-batch to its own `batch_id` directory makes replays idempotent. The claimed guarantee is effectively-once output per offset range; end-to-end exactly-once from the government source is *not* claimed. Rolling features and forecasts are recomputed only by the batch layer. This trade-off keeps a single feature implementation.

Storage. A three-DataNode HDFS cluster (replication 3, 128 MB blocks) holds the same layers under `/agri/{raw, bronze, silver, features, models, predictions, gold, checkpoints, metadata}`. Directories are partitioned only by low-cardinality columns (year, horizon) to avoid creating tens of thousands of small files. There is a single NameNode, so the store is not highly available.

8 Results

All numbers in this section are copied from files under `results/forecast/fc_main_v3/` and `results/anomaly/`. The test period (origins 2025-01-01 to 2026-04-21) was scored once, after every choice had been fixed on validation data. It contains about 1.55–1.67M forecasts per horizon.

8.1 Model selection on validation

On 2024 validation data, the absolute-loss GBT was selected for the next-report and 3-day horizons, and the random forest for 7 and 14 days.

At the next-report horizon:

- persistence scored a validation MAE of 210.7 INR/q;
- the best squared-loss learner (random forest) scored 211.1;
- Huber regression scored 212.5;
- the best absolute-loss GBT scored 205.3.

Under the original squared-loss-only search space, persistence would have been selected for this horizon. The other horizons would have been unchanged, except that 3 days would have selected the random forest.

Table 2: Test results, 2025-01-01 to 2026-04-21 (INR/quintal). Rel. = MAE / MAE(persistence) on the same rows. "Selected" is the family chosen on 2024 validation data.

Horizon	Model	MAE	RMSE	WAPE %	MASE	Rel.
next	Persistence	189.0	498.8	5.72	1.520	1.000
	Moving average 7 d	222.3	491.0	6.73	1.899	1.176
	Seasonal naive	965.7	1841.1	29.25	11.24	5.111
	Ridge (MLlib)	189.3	459.5	5.73	1.744	1.002
	Random forest	188.4	444.2	5.71	1.658	0.997
	GBT, absolute loss (sel.)	**183.5**	455.5	**5.56**	1.559	**0.971**
3 d	Persistence	242.6	574.8	7.37	1.716	1.000
	Random forest	232.5	509.9	7.06	1.783	0.958
	GBT, absolute loss (sel.)	**231.1**	521.1	**7.02**	1.680	**0.953**
7 d	Persistence	308.8	671.7	9.38	1.257	1.000
	Moving average 7 d	316.6	649.0	9.61	1.337	1.025
	GBT, absolute loss	291.7	612.0	8.86	1.312	0.944
	Random forest (sel.)	**289.7**	595.7	**8.80**	1.430	**0.938**
14 d	Persistence	397.0	812.0	12.07	1.113	1.000
	GBT, absolute loss	370.3	717.8	11.26	1.083	0.933
	Random forest (sel.)	**366.4**	717.1	**11.14**	1.056	**0.923**

8.2 Point forecasts

Table 2 reports the test results. The selected model has the lowest MAE at every horizon. Its MAE relative to persistence on identical rows is 0.971, 0.953, 0.938 and 0.923 for the next report and for 3, 7 and 14 days. Diebold–Mariano tests on daily mean absolute-error differences reject equal accuracy at every horizon. The HLN statistics range from -10.3 to -16.3, with $p < 10^{-8}$.

The gains are modest but broad:

- the selected model beats persistence on 58.5%, 68.8%, 75.0% and 76.4% of series with at least 30 test rows;
- at 7 days it improves on persistence for all 36 commodities, with relative MAE from 0.879 (jowar) to 0.969 (banana) (Figure 2).

The baselines show how hard this target is:

- seasonal naive is the worst forecaster (relative MAE 2.5–5.1, bias above +390 INR/q), because year-over-year level changes dominate;
- seasonal-adjusted persistence and the 30-day moving average are also worse than persistence;
- squared-loss ridge regression does not differ from persistence at the next-report horizon ($p = 0.47$).

MASE values above one for persistence (1.52 at the next-report horizon) show that test-period changes are larger than the training-period naive errors used as the scale.

A per-series statistical baseline. On a random sample of 400 panel series, we fitted simple exponential smoothing per series, with α chosen on training origins. Its relative MAE (0.975 and 0.962 at the next-report and 3-day horizons) matches the global model on the same rows (0.976 and 0.960). The global model is clearly better at 7 and 14 days (0.946 vs. 0.967, and 0.935 vs. 0.975). *At the shortest horizons, the advantage of machine learning over a well-tuned smoother is therefore nil.* It appears only once cross-market and seasonal information has time to matter.

Sub-periods. Before the AGMARKNET 2.0 migration (2025-01 to 2025-10), relative MAE is 0.970, 0.951, 0.937 and 0.922. After it (2025-11 to 2026-04, about 4–6% of test rows), the gains shrink to 0.993, 0.977, 0.968 and 0.944.

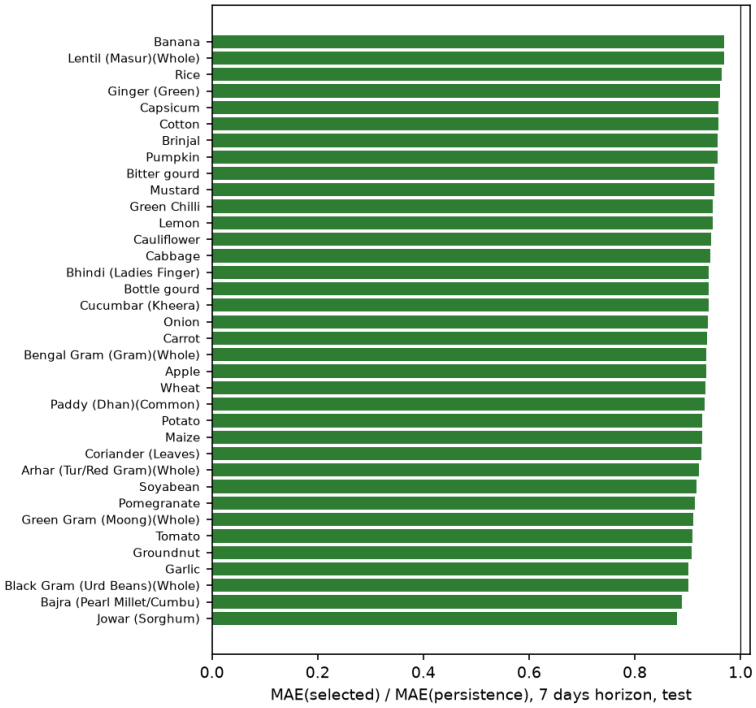


Figure 2: Test MAE of the selected model relative to persistence, by commodity, at the 7-day horizon.

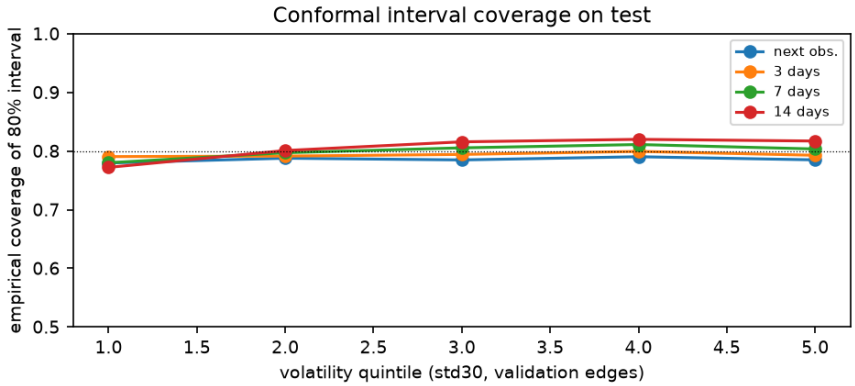


Figure 3: Test coverage of 80% split-conformal intervals by volatility quintile, with bin edges from validation.

Ablation. For the 7-day GBT, removing any one of the calendar, level or seasonal feature groups changes the test relative MAE by at most 0.004 (from 0.944). Removing the same-day cross-market features raises it to 0.960, and a returns-only model (three features) reaches 0.970. The most important random-forest features at 7 days are the 7-day deviation (0.18), the last return (0.12) and the national and state cross-market deviations (0.10 and 0.09). Information from other markets, not calendar or seasonality, is therefore what the models add over persistence.

8.3 Prediction intervals

Empirical test coverage of the nominal 80% intervals is 78.6%, 79.4%, 80.0% and 80.5%; for the 90% intervals it is 88.9–90.5%. Across volatility quintiles, coverage stays between 77.2% and 82.0% (Figure 3). Widths adapt strongly: the mean 80% width runs from 2–8% of the price in the calmest quintile to 45–88% in the most volatile. After the migration, however, 80% coverage falls to 67.6–72.9%. This is the failure mode under distribution shift that adaptive conformal methods address [7]; we did not implement them.

8.4 Decision rule

On the test period, the rule abstains (WATCH) on 97.8–98.5% of report-days. At 7 days:

- HOLD fires on 0.5% of report-days; the realised price beats today's price plus holding cost 80.0% of the time;
- SELL fires on 1.0%; the realised price is below today's 67.8% of the time.

The median realised 7-day change after HOLD is +46.8%. This reveals what the rule mostly detects: reports far below the series' recent level that subsequently revert. Some of these may be reporting errors smaller than our factor-five sanity threshold. These are hit rates on published modal prices, not farm-gate profits, and COMPARE could not be back-tested because no transport data was available.

8.5 Anomaly detection

On 400k real test rows with 1% injected corruptions, the production rule reaches PR-AUC 0.704 for any corruption. This compares with 0.671 for a previous-report change score and 0.352 for the legacy commodity-IQR rule. At its operating point, precision is 0.31, recall 0.85 and the false-positive rate 1.9%. For pure unit or keying errors the simple change score ranks best (PR-AUC 0.996). Abrupt ±35–60% shocks are hard for every detector (best PR-AUC 0.25). The rule flags 1.9% of uncorrupted real rows; without labels we cannot say how many of these are genuine market events.

8.6 Stability over time

A rolling-origin backtest refits each horizon's selected configuration on expanding windows and tests on 2021, 2022, 2023 and 2024. The resulting relative MAE ranges are:

- next report: 0.970–0.976;
- 3 days: 0.952–0.960;
- 7 days: 0.936–0.946;
- 14 days: 0.920–0.936.

These bracket the main test values, so the improvement over persistence is not an artefact of one test year. The 2024 fold coincides with the validation year.

8.7 Streaming

We replayed 20,000 real archived reports through a six-partition Kafka topic at 2,000 events/s with labelled faults: 5% duplicates, 2% late re-sends, 1% malformed messages, and local reordering. Of 21,524 messages:

- 20,000 distinct events reached silver, with none missing and none never sent;
- 1,317 duplicate or late copies were removed by watermarked de-duplication;
- 207 malformed messages were quarantined;
- none were dropped by the watermark.

The canonicalised streaming rows matched the batch canonicalisation of the same records in all 14 compared columns, with 0 mismatches. The median per-batch producer-to-sink latency was 1.7 s with a 2 s trigger. The red-team review found that an earlier version let a malformed message advance the event-time watermark and silently discard every later event. Unusable messages are now quarantined before the watermark, and a regression test covers this case.

8.8 Storage and distributed execution

The silver layer (692 MB, 1,743 files) was loaded into a three-DataNode HDFS with replication 3. fsck reported no missing or corrupt blocks, and a file remained readable, with an identical checksum, while a DataNode was stopped. All DataNodes share one physical disk, and there is a single NameNode. On a containerised standalone Spark cluster (2 workers, 6 cores, 1 GB executors) reading from HDFS, the aggregation workload processed 11.3M real rows in 9.2 s and the window-feature workload in 14.4 s. These runs exercise the distributed code path (separate driver and executor JVMs, network shuffle, HDFS reads) but not multi-machine scaling.

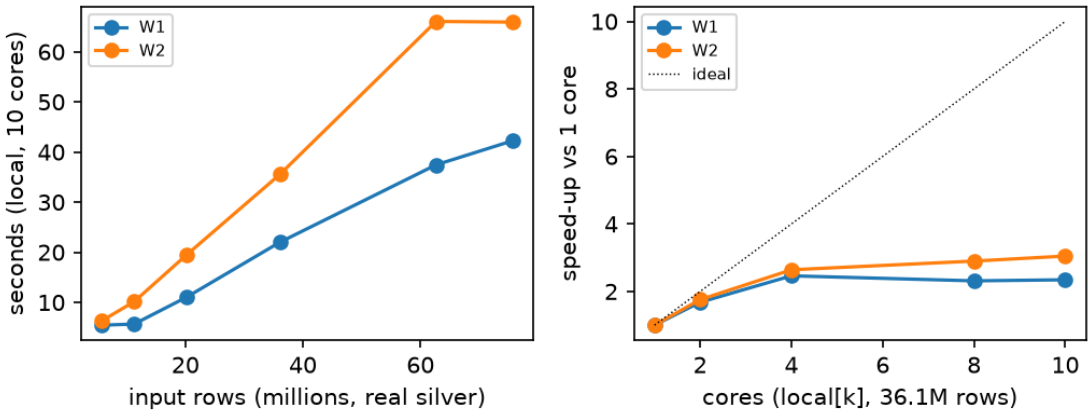


Figure 4: Left: run time against input size on the full silver layer (local mode, 10 cores). Right: speed-up against core count for 36.1M rows. W1 is an aggregation; W2 is per-entity window features.

8.9 Single-host scalability

On one Apple M5 laptop (4 performance and 6 efficiency cores, 16 GB), run time grows roughly linearly with input up to the full 75.6M-row silver layer (Figure 4). Throughput is 1.6–2.0M rows/s for the aggregation and about 1.0M rows/s for the window features, with about 1 GB of shuffle at full size and no disk spill. Speed-up with cores flattens after four, matching the performance-core count: 2.5× for the aggregation and 3.0× for the window features at 10 cores. The complete feature build for the forecasting panel (33M reports, 25 features, four targets) takes 8.3 minutes (`logs/f1_build_features.log`), and the full benchmark (38 model fits on 2M rows each, i.e. 32 candidates and 6 ablations, plus scoring of 6.4M test forecasts) takes 1 h 41 min.

8.10 Comparison with published results

No published study we found uses daily mandi modal prices with these horizons, chronological purged splits and baselines. We therefore quote prior numbers only as context.

- Manogna et al. [20] report a GRU MAPE of 14.59% for onion and 10.58% for tomato in their abstract, and 16.17% and 14.81% in the body. These come from an 80/20 hold-out of unclear horizon and aggregation. The authors explicitly chose not to use MASE, and we found no naive baseline in the sections we could read.
- Paul et al. [25] evaluate 17 Odisha markets for brinjal on a 90/10 split.
- Paul et al. [26] use monthly data.

Our WAPE of 5.6–11.1% is not comparable with these MAPE figures: the denominators, horizons, aggregation and series differ. The comparison we can defend is internal. On our data and splits, tuned learners beat persistence by 3–8% in MAE, and beat a per-series smoother only beyond three days. This is far smaller than the gains over ARIMA reported in the literature, for example GRU RMSEs four to six times lower than ARIMA's for onion and tomato in [20]. The difference suggests that the choice of baseline and protocol, rather than the choice of model, explains much of the spread in published results.

9 Farmer application and decision support

Design constraints. The intended users have entry-level Android phones (about 4 GB of RAM), limited data plans and varied literacy. The application is a progressive web app written in plain JavaScript modules, with no framework, CDN or build step:

- 69 KB of JavaScript in total, of which about 29 KB loads for the first screen; the chart and the other screens are loaded lazily;
- a service worker precaches the application shell and fetches API data network-first with a timeout;
- when offline, it falls back to cached responses and labels them with the time they were saved.

All 264 interface strings, including decision reasons, warnings, errors, accessibility labels and a ten-topic farmer guide, exist in English, Tamil, Telugu, Malayalam, Hindi and Kannada. Automated tests enforce identical key sets, identical placeholders and native-script content in every language. Commodity names are always shown in their original published form, with a localised label from a reviewed dictionary where one exists. The translations are AI-drafted and *provisional* until reviewed by native speakers with agricultural vocabulary.

Freshness and provenance. Every price shows its report date and age. A banner appears whenever the dataset is more than three days old. Because our deployment serves an archive that ends on 2026-04-21, it says that the data is a historical snapshot and never calls it "live". Forecast screens show the model version, the source and licence, the measured test error for that commodity and horizon, and the measured coverage of the interval.

Decision cards. The cards turn the 80% interval $[\ell_h, u_h]$ (on the log-return scale) into a transparent rule. The holding cost is $c_h = c_0 + w \cdot h/7$: a fixed 1% plus 5% per week for perishables, or 1% per week for durables. These costs are placeholders that the farmer can edit.

- HOLD if $\ell_h \geq \ln(1 + c_h)$;
- SELL if $u_h \leq 0$;
- COMPARE if a market in the same district reported a price at least 5% higher within three days;
- WATCH otherwise.

Every card is labelled a *market outlook*, not a profit recommendation, and has a "Why am I seeing this?" panel listing the interval, costs, accuracy and freshness. The client recomputes the card when the farmer enters transport or holding costs. The rule's test-period hit rates are reported in Section 8.4.

10 Threats to validity

- **Internal.** Hyper-parameters, model family and conformal quantiles were all chosen on 2024 data. The search space was amended once, after a validation-only observation (Section 6). The panel was selected on 2021–2023 reporting density. For backtest folds inside that window, this selection uses within-fold information about which series report often, but no prices.
- **External.** Results describe well-reported mandis of 36 commodities. The test period includes a change in the publishing system, and the source is a third-party mirror of official data.
- **Construct.** Modal price is the published summary statistic, not the price an individual farmer received. Exactly-zero changes may partly reflect reporting practice rather than market stability.
- **Statistical.** Diebold–Mariano tests treat daily cross-sectional mean losses as one time series. We report one test per horizon and model against persistence, without a multiplicity correction; p-values are descriptive. The Newey–West lag equals the far edge of the target window ($h + 3$; 7 for the next report). Doubling it to 20 at the 7-day horizon still gives $p \approx 10^{-32}$.

11 Ethical considerations

- **Harm from false certainty.** Forecast errors can hurt farmers who act on them. The interface therefore shows intervals, measured error and data age, never guarantees a price, and offers WATCH by default.
- **Privacy.** No personal data is collected, and there are no accounts or analytics.
- **Licensing.** The data is used under GODL-India with attribution.
- **Language.** Machine-drafted translations are marked provisional, because a mistranslated warning is itself a harm.
- **Evidence on information services.** Information services do not reliably raise farm-gate prices [5, 23]. Welfare claims would require a field study, which we have not done.

12 Reproducibility

All code, configuration and the experiment registry are in the repository. The following are deterministic given the archive:

- the silver, feature, model, prediction and gold layers;
- every table and figure in this paper.

These are produced by `scripts/run_forecast_pipeline.sh` and the analysis modules under `py/agri/analysis`. Each experiment writes a manifest recording the git commit, the Spark and Java versions, the seed, the periods, row counts, the search space, the selected models and any warnings. The test suites cover leakage, metric definitions, split purging, conformal calibration, the decision rule, batch/stream parity, de-duplication, late events, checkpoint recovery, the ingestion adapter, API contracts and internationalisation. The raw archive is not redistributed and can be downloaded from its public source.

13 Limitations

- **Archived data.** The system runs on an archive that ends on 2026-04-21. The live API adapter has been tested only against fixtures, because no key was available.
- **Training and coverage of the panel.** Models were trained once on data up to 2023. The panel favours well-reported mandis, so thinly reported markets receive forecasts of unknown accuracy.
- **Features.** Same-day cross-market features assume timely publication of other markets' reports. They are also the most useful feature group, so this assumption matters.
- **Intervals.** Coverage degrades under the source change, and we did not implement adaptive conformal inference.
- **Decision support.** COMPARE is not back-tested.
- **Infrastructure.** All distributed runs used one physical machine (local mode, and a containerised standalone cluster). There is a single NameNode and a single Kafka broker.
- **Application.** Interface translations are AI-drafted and provisional. No user study and no test on a physical low-end phone was carried out.

14 Conclusion and future work

On a national panel of Indian mandi series evaluated with a leakage-resistant protocol, learned models improve on persistence by 3–8% in MAE. They improve on a per-series smoother only beyond three days. Most of the added skill comes from same-day cross-market information. Conformal intervals are close to nominal until the data source changes. These results are less dramatic than many published gains, and we argue they are a more realistic basis for farmer-facing tools.

Future work, in order of expected value:

1. live ingestion with a data.gov.in key, followed by monthly retraining evaluated with rolling origins;
2. adaptive conformal intervals that recover coverage after source changes;
3. arrivals and weather covariates;
4. native-speaker review of all translations, and a field usability study with farmers;
5. a transport-cost model so that COMPARE can be evaluated;
6. a multi-machine cluster for scalability evidence beyond one host.

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